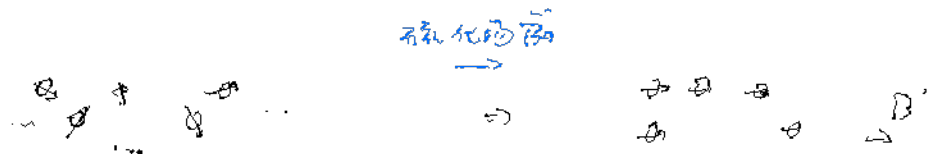


第七章 磁介质.

§ 1 分子电流观点

1. 图像

单个分子 $\vec{I}$ $\vec{a}$ $\leftrightarrow$ 小圆线圈 线圈 $\leftrightarrow$ 磁偶极层.
 $\vec{m} = I \vec{a}$

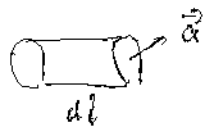


2. 磁化强度 $\vec{M}$

$$\vec{M} = \sum_{i=1}^N \vec{m}_i$$

$$= n \vec{m} \quad (\text{假设各分子相同, 数密度为 } n)$$

在面 S 边界为 L 取 $d\vec{l}$, 底面为 $\vec{a}$



$$dV = \vec{a} \cdot d\vec{l}$$

dV 内分子电流环都穿过 S (一次)

$$d\Phi = I \cdot (n \cdot \vec{a} \cdot d\vec{l}) = n \vec{m} \cdot d\vec{l}$$

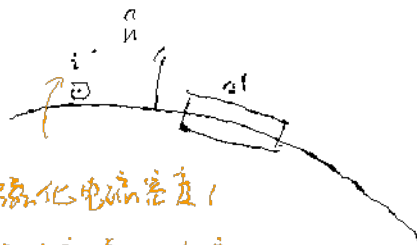
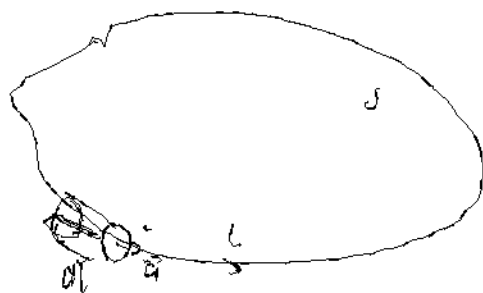
$$= \vec{M} \cdot d\vec{l}$$

$$\Rightarrow \oint_L \vec{M} \cdot d\vec{l} = \sum I' \quad \text{穿过 } S \text{ 的分子电流.}$$

在表面附近回路.

$$\oint_L \vec{M} \cdot d\vec{l} = n \vec{m} \cdot d\vec{l} = \sum I' = i' dl$$

$$\Rightarrow \vec{M} \times \vec{n} = \vec{i}' \quad \text{数值上 } i' = M$$



面磁化电流密度

单位长度磁化电流

3. (磁介质中的) 磁感应强度 $\vec{B}$

有磁介质时: $\vec{B} = \vec{B}_0 + \vec{B}'$

例: 沿轴 均匀磁化 的磁介质圆柱.

$$\vec{M} = M \hat{z} \quad \Rightarrow \quad \text{变化磁化率}$$

$$B = \frac{\mu_0 n^2}{2} (\cos \beta_1 - \cos \beta_2) \quad (n \text{ 为单位长度内线圈匝数})$$

$$\begin{aligned} \hookrightarrow B' &= \frac{\mu_0 M}{2} (\cos \beta_1 - \cos \beta_2) = \frac{\mu_0 M}{2} (\cos \beta_1 - \cos \beta_2) \quad \cos \beta = \frac{z}{\sqrt{z^2 + R^2}} \\ &= \mu_0 M \frac{z}{\sqrt{z^2 + R^2}} = \begin{cases} \mu_0 M & \text{无限长 } \frac{1}{R} \rightarrow \infty \\ 0 & \text{薄片 } \frac{1}{R} \rightarrow 0 \end{cases} \quad B = B_0 + \mu_0 M \end{aligned}$$

4. 磁场强度矢量

$$\begin{aligned} \oint_L \vec{B} \cdot d\vec{s} &= \mu_0 \oint \vec{J} \cdot d\vec{s} \\ &= \mu_0 \oint (\vec{J}_0 + \vec{J}') \cdot d\vec{s} \\ &= \mu_0 \oint \vec{J}_0 \cdot d\vec{s} + \mu_0 \oint \vec{J}' \cdot d\vec{s} \end{aligned}$$

引入磁场强度矢量 H

$$\vec{H} = \frac{\vec{B}}{\mu_0} - \vec{M} \quad \text{在真空中 } M=0, \vec{B} = \mu_0 \vec{H}$$

$$\Rightarrow \oint_L \vec{H} \cdot d\vec{l} = \oint \vec{J}_0 \cdot d\vec{s} = I_0 \quad / \quad \nabla \times \vec{H} = \vec{J}_0$$

例: 计算无限磁介质的螺绕环内的磁感应强度 B , 已知磁化强度 M

$$\oint \vec{H} \cdot d\vec{l} = 2\pi r H = I_0$$

$$B_0 = \mu_0 n I_0 = \mu_0 H \quad (\text{磁化率为真空螺绕环})$$

$$B = \mu_0 (H + M) = B_0 + \mu_0 M, \text{ 与无限长直 } \sim \text{ 一致}$$

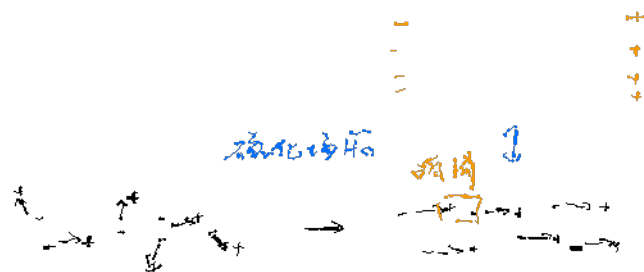
§2 磁荷观点.

磁序会定得: $\vec{F} = \frac{1}{4\pi\mu_0} \sum_{i,j} \frac{q_{mi} q_{mj}}{r_{ij}^2} \hat{r}_{ij}$

小磁针 $\hookrightarrow$ 电偶极子

1. 磁矩化矢量 $\vec{J}$

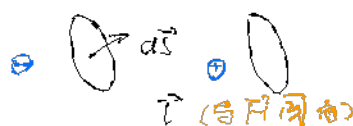
$$\vec{J} := \frac{\sum \vec{p}_m}{2eV}$$



2.

$$\vec{p}_m = q_m \vec{l} \rightarrow \vec{J} = n \vec{p}_m = n q_m \vec{l}$$

取如图所示面:



极化的第 i 面 dS 的极化磁荷为

$$n q_m \vec{l} \cdot d\vec{S} = \vec{J} \cdot d\vec{S}$$

$$\hookrightarrow \oint \vec{J} \cdot d\vec{S} = - \sum_{i,j} q_{mj}$$

$$\oint \vec{J} \cdot d\vec{S} = - \oint d\vec{S} \cdot \vec{n} + 0$$

$$= - 6\omega dS$$

$$\Rightarrow 6\omega = \vec{J} \cdot \vec{n}$$

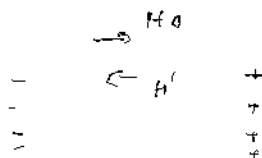


3. 退磁场 / 退磁因子

$$\vec{H} = \vec{H}_0 + \vec{H}'$$

在介质内部:

$$H = H_0 - H' \quad (\text{退磁场})$$



分析:

$$H' \sim 6\omega = J$$

$$\sim l \cdot d \rightarrow \sim \frac{l}{d} \quad (\text{几何相似性})$$

$$H' = \frac{1}{\mu_0} N_0 J \quad \text{其中 } N_0 = N_0(\frac{l}{d}), \text{ 为退磁因子}$$

例: 根据磁荷定律计算均匀磁化的圆盘的退磁场 H' ,

[磁极化强度], 棒长 l , 直径 d

$$b_m = J$$

$$-b - 0 \rightarrow J \quad 0 b_m$$

在中点, 两个面.

$$dH' = 2 \cdot \frac{1}{4\pi\mu_0} \cdot \frac{b_m \cdot 2\pi r dr}{[r^2 + (\frac{l}{2})^2]^{3/2}} \cdot \frac{l}{2}$$

$$= 2 \cdot \frac{b_m l}{4\pi\mu_0} \cdot \frac{r dr}{[r^2 + (\frac{l}{2})^2]^{3/2}}$$

$$H' = 2 \int_0^R dH' = 2 \cdot \frac{b_m l}{4\pi\mu_0} \left(r^2 + (\frac{l}{2})^2 \right)^{-3/2} \Big|_0^R$$

$$= 2 \cdot \frac{J l}{4\pi\mu_0} \left(\frac{2}{l} - \frac{1}{\sqrt{R^2 + (\frac{l}{2})^2}} \right)$$

$$= \frac{J l}{\mu_0} \left(\frac{1}{l} - \frac{1}{\sqrt{l^2 + d^2}} \right) = \frac{J}{\mu_0} \left(1 - \frac{1}{\sqrt{1 + (\frac{d}{l})^2}} \right) J$$

$$\Rightarrow N_D = \left(1 - \frac{1}{\sqrt{1 + (\frac{d}{l})^2}} \right) = \begin{cases} 0 & \text{无限长磁棒 } \frac{d}{l} \rightarrow \infty \\ 1 & \text{薄片 } \frac{d}{l} \rightarrow 0 \end{cases}$$

无限长磁棒 ~ 闭合磁路:

$$H' = 0, \text{ 周磁化}$$

4. 安培环路定理 / 高斯定理

$$H_0 = \frac{1}{4\pi} \oint_{(L)} \frac{2 \cdot d\vec{r}_i \times \vec{r}_i}{r_i^3} \quad (H \text{ 与 } B \text{ 类似})$$

$$\rightarrow \oint H_0 \cdot d\vec{l} = \frac{2}{c} I_0$$

$$\oint H_0 \cdot d\vec{S} = 0$$

$$\vec{H}' = \frac{1}{4\pi\mu_0} \int \frac{d\vec{q}_m}{r^2} \vec{r} \quad (H' \text{ 与 } E \text{ 类似})$$

$$\rightarrow \oint \vec{H}' \cdot d\vec{l} = 0$$

$$H' \cdot \vec{r}_i \cdot \vec{r}_i = 1, 2, 3, 4$$

$$\vec{H} = \vec{H}_0 + \vec{H}'$$

$$\Rightarrow \begin{cases} \oint \vec{H} \cdot d\vec{l} = \sum I' \\ \oint \vec{H} \cdot d\vec{s} = \frac{1}{\mu_0} \sum q_m \end{cases}$$

5. 磁感应强度矢量 $\vec{B}$

$$\vec{D} = \epsilon_0 \vec{E} + \vec{P}$$

$$\vec{B} = \mu_0 \vec{H} + \vec{J}$$

$$\Rightarrow \oint \vec{B} \cdot d\vec{s} = \mu_0 \oint \vec{H} \cdot d\vec{s} = \sum q_m = 0$$

6. 磁化率和磁导率

$$\vec{P} = \chi_e \epsilon_0 \vec{E}$$

$$\vec{J} = \chi_m \mu_0 \vec{H}$$

$$\vec{B} = \mu_0 \vec{H} + \vec{J} = (1 + \chi_m) \mu_0 \vec{H} = \mu_r \mu_0 \vec{H}$$

例：求绕在磁导率为 μ 的闭合磁环上的载流线圈与同轴匝数、尺寸

的空心螺线管的自感之比

$$H = \frac{N}{2\pi r} I_0 = n I_0$$

$$\frac{B}{B_0} = \frac{\mu_r \mu_0 H}{\mu_0 H} = \mu_r$$

$$\frac{L}{L_0} = \frac{\Psi}{\Psi_0} = \frac{B}{B_0} = \mu_r$$

§4 两种观点的等效性.

电偶层 ~ 磁偶极子

考虑 - 个电偶极子微元在 P 处产生电势

$$u(p) = \frac{q_0 d \cos \theta}{4\pi\epsilon_0} \left(\frac{1}{r'} - \frac{1}{r} \right)$$

$$r' \approx r + l \cos \theta$$

$$\frac{1}{r'} = \frac{1}{r + l \cos \theta} = \frac{1}{r} \frac{1}{1 + \frac{l \cos \theta}{r}} \approx \frac{1}{r} \left(1 - \frac{l \cos \theta}{r} \right)$$

$$\rightarrow u(p) = \frac{q_0 d \cos \theta}{4\pi\epsilon_0} \left(\frac{1}{r} - \frac{l \cos \theta}{r^2} \right) = - \frac{q_0 d \cos \theta}{4\pi\epsilon_0} \frac{\vec{r} \cdot \vec{r}}{r^3}$$

$$= - \frac{1}{4\pi\epsilon_0} \frac{\vec{p}_e \cdot \vec{r}}{r^3}$$

$$= - \frac{q_0 l}{4\pi\epsilon_0} \frac{d\vec{s} \cdot \vec{r}}{r^2}$$

$$= - \frac{\tau_e}{4\pi\epsilon_0} \cdot d\Omega$$

面电偶极矩密度.

$$p_e = q_0 d s$$

$$\tau_e = q_0 l \quad d\Omega = \frac{d\vec{s} \cdot \vec{r}}{r^2} = \frac{d\vec{l} \cdot \vec{r}}{r^2}$$

等效磁荷理论:

$$\vec{F} = \frac{1}{4\pi\mu_0} \frac{Q_m Q_m}{r^3} \vec{r}$$

$$\vec{H} = \frac{\vec{F}}{Q_m} = \frac{Q_m}{4\pi\mu_0} \frac{\vec{r}}{r^3}$$

$$\oint \vec{H} \cdot d\vec{l} = 0 \rightarrow H = -\nabla u_m \quad (\text{对 } u_m \text{ 保守势})$$

对 j 磁偶极子 $\vec{p}_m$

$$u_m = \frac{1}{4\pi\mu_0} \frac{\vec{p}_m \cdot \vec{r}}{r^3}$$

$$p_e = q_0 l a s$$

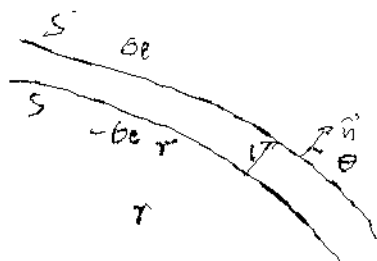
$$\tau_m = q_0 l \cdot l$$

$$= \frac{\tau_m}{4\pi\mu_0} d\Omega$$

$$H = -\nabla u_m = \frac{\tau_m}{4\pi\mu_0} \nabla \Omega \sim \vec{B} = \frac{\mu_0}{4\pi} \nabla \Omega$$

$$\Rightarrow \vec{B} = \mu_0 \vec{H}$$

$$p_m = \mu_0 \vec{H}$$



§ 5 磁介质的磁化规律和机理 铁电体.

磁矩: 轨道磁矩.

$$\frac{e}{4\pi\epsilon_0 r^2} = m \frac{v^2}{r} \rightarrow v = \sqrt{\frac{e^2}{4\pi\epsilon_0 m r}}$$

$$I = \frac{e}{T} = \frac{e v}{2\pi r}$$

$$\mu_L = IS = 2\pi r^2 = \frac{e v}{2} r = \frac{e^2}{4} \sqrt{\frac{r}{2\pi\epsilon_0 m}}$$

$$\frac{1}{2} \sigma_L^2 \frac{e}{2m} \quad L = m v r$$

$$\vec{\mu}_L = -\gamma \vec{L} \Rightarrow \text{轨道磁矩是量子化的}$$

角动磁矩 $\sim$ 角动

$$\text{角动角动量 } L_s = \frac{h}{2\pi}$$

$$\mu_s = -\frac{e}{m} L_s = -\frac{e h}{4\pi m}$$

$$\frac{1}{2} \sigma_s = \frac{e}{m} \quad \text{Landé 因子 } g := \frac{\sigma_s}{\sigma_L} \approx 2.0023$$

$$\mu_s = g \mu_B L_s$$

$\sim$ 铁磁共振

$$\vec{J} = \chi_m \mu_0 \vec{H} \quad \chi_m \begin{cases} > 0 & \text{顺磁介质} \\ < 0 & \text{抗磁介质} \end{cases}$$

$$B = \vec{B}_0 + \vec{B}_m \quad \text{放入介质} \begin{cases} B > B_0 & \text{顺磁介质} \\ B < B_0 & \text{抗磁介质} \\ B \gg B_0 & \text{电磁介质} \end{cases}$$

经典观机版：

1) $\vec{r}$ 为固有磁矩 (顺磁效应)

$$\vec{m} = 2s \vec{r} = \frac{-e}{2m} \vec{r}^2, \quad \vec{r}^2 = -\frac{e \hbar^2}{2m} \vec{L}$$

$B = 0$: 热运动 $\rightarrow$ 取向无规, $\sum \vec{m} = 0$

$B \neq 0$: $\vec{m}$ 按 $\vec{B}$ 方向排列.

$\Rightarrow$ 顺磁.

2) $\vec{r}$ 为无固有磁矩 (抗磁效应)

$$\frac{Ze^2}{4\pi\epsilon_0 r^2} = m \omega^2 r \quad \rightarrow \quad \omega = \sqrt{\frac{Ze^2}{4\pi m \epsilon_0 r^3}}$$

$$\begin{matrix} \uparrow \omega_0 \\ \uparrow \vec{B} \\ Ze \\ (+) \end{matrix} \quad \ominus e$$

假设加上外场 $\vec{B}$ ($B \ll \frac{m \omega_0^2}{e}$) 时, 轨道半径不变.

$$\frac{Ze^2}{4\pi\epsilon_0 r^2} + e \hbar B r = m \omega^2 r \quad \omega^2 = (\omega_0 + \Delta\omega)^2 \approx \omega_0^2 + 2\omega_0 \Delta\omega$$

$$\Rightarrow \Delta\omega = \frac{e \hbar B r}{-e B r + 2m \hbar \omega} \approx \frac{e \hbar B}{2m}$$

$$\vec{m} = -\frac{e \hbar^2}{2m} \vec{L}$$

$$\Rightarrow \Delta \vec{m} = -\frac{e \hbar^2}{4\pi m} \vec{B}$$

(实际 $\vec{B}$ 上除有外场由电对占据, 但 $\vec{B}$ 方向不影响 $\Delta \vec{m}$ 方向)

$$\begin{aligned} \langle \Delta m \rangle &= -\frac{e \hbar^2}{4\pi m} \cdot \langle L^2 \rangle \quad \text{球坐标} \quad \langle r^2 \rangle = \frac{1}{4\pi r^2} \int_0^\pi \int_0^{2\pi} (\rho \sin\theta)^2 \rho^2 \sin\theta d\theta d\varphi \\ &= -\frac{e \hbar^2}{4\pi m} \frac{1}{\vec{B}} = \frac{e^2 \hbar^2}{4\pi m} \int_0^\pi \sin^3\theta d\theta \int_0^{2\pi} d\varphi = \frac{2}{3} \rho^2 \quad \vec{B} \end{aligned}$$

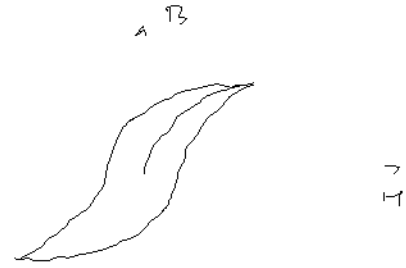
倘若 $\vec{B}$ 与 $\vec{H}$ 方向不同, 也可以证明 $\vec{B}$ 与 $\vec{H}$ 方向相反

一般情况下如何证明:

铁磁质:

磁滞回线:

$$\vec{B} = \mu(\vec{H}) \cdot \vec{H}$$



能量耗散:

线圈感生电动势, $\mathcal{E} = -\frac{d\psi}{dt}$

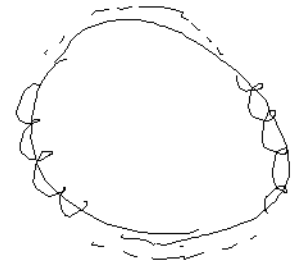
电源做功

$$dQ = -I \mathcal{E} dt = I d\psi$$

$$= \frac{H}{N} d(N\psi) = \frac{H}{N} L \cdot N S \cdot dB = SL H dB$$

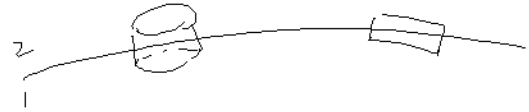
$$Q = \oint \frac{dB}{N} = H \oint dB$$

$\Rightarrow$ 磁滞回线所围面积 = 单位面积铁芯损耗能量.



§ 6 电磁介质界面上的边界条件 磁路定理

$$\begin{cases} \oint \vec{B} \cdot d\vec{S} = 0 \\ \oint \vec{H} \cdot d\vec{l} = \sum I_c = 0 \end{cases} \quad \text{在界面上}$$



$$\begin{cases} \oint \vec{B} \cdot d\vec{S} = (B_{2n} - B_{1n}) dS = 0 \\ \oint \vec{H} \cdot d\vec{l} = (H_{2\tau} - H_{1\tau}) dl = 0 \end{cases}$$

$$\Rightarrow (\vec{B}_2 - \vec{B}_1) \cdot \hat{n} = 0, (\vec{H}_2 - \vec{H}_1) \times \hat{n} = 0$$

$$\begin{matrix} B_{1n} & = & B_{2n} \\ H_{1\tau} & = & H_{2\tau} \end{matrix}$$

$$\hookrightarrow \frac{\mu r_1}{\tan \theta_1} = \frac{\mu r_2}{\tan \theta_2}$$

磁路:

$$\Phi = BS = \text{const.}$$

$$\oint \vec{H} \cdot d\vec{l} = NI$$

$$\hookrightarrow \sum \oint \vec{H} \cdot d\vec{l} = NI$$

$$\Rightarrow \begin{cases} \text{磁阻} & \frac{l}{\mu r \mu_0} & \sim & R \\ \Phi & & \sim & \frac{1}{R} \\ \text{磁通势} & NI & \sim & U \end{cases}$$

在气隙 ($\mu_r \approx 1$) 处 R_m 远大于其它处

$\Rightarrow$ 有气隙时, 回路中 Φ 很小, $B \approx 0$

磁屏蔽:

空腔与金属壁并联

$$R_{m\text{空}} \gg R_{m\text{壁}}$$

$$\rightarrow \Phi_{\text{空}} \ll \Phi_{\text{壁}} \rightarrow B_{\text{空}} \approx 0$$



可用多层铁壳来提升屏蔽效果.

例 7.2

磁介质: $B = \frac{1}{2} \mu_0 n i (\cos \beta_1 - \cos \beta_2)$

① $i = n$

内部: $B = \mu_0 n i = \mu_0 n$

$H_1 = \frac{B_1}{\mu_0} - n = 0$

$B_3 = B_4 = B_6 = B_7 = \frac{1}{2} \mu_0 n$

$H_5 = H_6 = -\frac{1}{2} n$

$B_2, B_5 = 0$

$H_4 = H_7 = \frac{1}{2} n$

$H_2 = H_3 = 0$

② $\Delta m = 1 = \mu_0 n$

$H = \frac{1}{4\pi\mu_0} \oint \frac{q_m \vec{r}}{r^2}$

或 $\oint \vec{H} \cdot d\vec{l} = I$

$H = \int_0^R \frac{1}{4\pi\mu_0} \frac{\Delta m \cdot 2\pi r \cdot dr \cdot X}{(r^2 + X^2)^{3/2}}$

$= \frac{\Delta m}{2\mu_0} \int_0^R \frac{X \cdot dr}{(r^2 + X^2)^{3/2}} = \frac{\Delta m}{2\mu_0} X \cdot \left. \frac{1}{(r^2 + X^2)^{1/2}} \right|_0^R$

$= \frac{\Delta m}{2\mu_0} \left(1 - \frac{X}{\sqrt{R^2 + X^2}} \right) = \frac{1}{2} n \left(1 - \frac{X}{\sqrt{R^2 + X^2}} \right)$

$H_1 = n \cdot \left[1 - \frac{\frac{L}{2}}{\sqrt{R^2 + \frac{L^2}{4}}} \right] \stackrel{R \ll L}{\approx} 0$

$H_4 = \frac{1}{2} n - \frac{1}{2} n \left(1 - \frac{\frac{L}{2}}{\sqrt{R^2 + \frac{L^2}{4}}} \right) = \frac{1}{2} n \frac{\frac{L}{2}}{\sqrt{R^2 + \frac{L^2}{4}}} \approx \frac{1}{2} n$

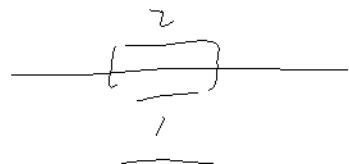
$H_5 = -\frac{1}{2} n - \frac{1}{2} n \left(1 - \frac{\frac{L}{2}}{\sqrt{R^2 + \frac{L^2}{4}}} \right) \approx -\frac{1}{2} n$

$H_6 = -\frac{1}{2} n - \frac{1}{2} n \left(1 - \frac{\frac{L}{2}}{\sqrt{R^2 + \frac{L^2}{4}}} \right) \approx -\frac{1}{2} n$

$H_7 = \frac{1}{2} n - \frac{1}{2} n \left(1 - \frac{\frac{L}{2}}{\sqrt{R^2 + \frac{L^2}{4}}} \right) \approx \frac{1}{2} n$

由安培环路定理

$H_2 = H_3 = H_1 = 0$



beg 7.11

$$r \begin{pmatrix} 1 \end{pmatrix} \times$$

$$U_m = \int_0^R \frac{1}{4\pi\epsilon_0} \frac{6m r^2 dr}{(r^2 + X^2)^{3/2}}$$

$$= \frac{6m}{2\mu_0} \int_0^R \frac{1}{(r^2 + X^2)^{3/2}} dr = \frac{6m}{2\mu_0} (R^2 + X^2 - X)$$

$$H = \frac{6m}{2\mu_0} \left(1 - \frac{X}{R^2 + X^2} \right) + \left(r - \frac{L}{2} \right) - \left(r + \frac{L}{2} \right)$$

$$U_m = \frac{6m}{2\mu_0} (R^2 + (r - \frac{L}{2})^2 + R^2 + (r + \frac{L}{2})^2 - L)$$

$$H = \frac{\partial U_m}{\partial r} \left(- \frac{r - \frac{L}{2}}{R^2 + (r - \frac{L}{2})^2} + \frac{r + \frac{L}{2}}{R^2 + (r + \frac{L}{2})^2} \right) = \frac{\partial U_m}{\partial r}$$

beg 7.15

$$H_1 = \frac{6m}{2\mu_0} \left(-1 + \frac{X_1 + \frac{L}{2}}{R^2 + (X_1 + \frac{L}{2})^2} - 1 + \frac{\frac{L}{2} - X_1}{R^2 + (\frac{L}{2} - X_1)^2} \right)$$

$$= \frac{6m}{2\mu_0} \left(\frac{X_1 + \frac{L}{2}}{R^2 + (X_1 + \frac{L}{2})^2} + \frac{\frac{L}{2} - X_1}{R^2 + (\frac{L}{2} - X_1)^2} - 2 \right)$$

$$H_2 = \frac{6m}{2\mu_0} \left(-1 + \frac{X_2 + \frac{L}{2}}{R^2 + (X_2 + \frac{L}{2})^2} + 1 - \frac{\frac{L}{2} - X_2}{R^2 + (\frac{L}{2} - X_2)^2} \right)$$

$$= \frac{6m}{2\mu_0} \left(\frac{X_2 + \frac{L}{2}}{R^2 + (X_2 + \frac{L}{2})^2} - \frac{\frac{L}{2} - X_2}{R^2 + (\frac{L}{2} - X_2)^2} \right)$$

$$B_1 = \mu_0 H_1 + J = \mu_0 H_1 + 6m$$

$$= \frac{6m}{2} \left(\frac{X_1 + \frac{L}{2}}{R^2 + (X_1 + \frac{L}{2})^2} + \frac{\frac{L}{2} - X_1}{R^2 + (\frac{L}{2} - X_1)^2} \right)$$

$$B_2 = \mu_0 H_2 = \frac{1}{2} 6m \left(\frac{X_2 + \frac{L}{2}}{R^2 + (X_2 + \frac{L}{2})^2} - \frac{\frac{L}{2} - X_2}{R^2 + (\frac{L}{2} - X_2)^2} \right)$$

等效磁偶：

$$B_c' = \frac{1}{2} \mu_0 n I \cdot (\cos \theta_1 - \cos \theta_2)$$

$$= \frac{1}{2} \mu_0 n I \left(\frac{x_1 + \frac{l}{2}}{\sqrt{R^2 + (x_1 + \frac{l}{2})^2}} + \frac{\frac{l}{2} - x_1}{\sqrt{R^2 + (\frac{l}{2} - x_1)^2}} \right)$$



$$B_m = J = \mu_0 m = \mu_0 n I$$

$$= \frac{1}{2} B_m \left(\frac{x_1 + \frac{l}{2}}{\sqrt{R^2 + (x_1 + \frac{l}{2})^2}} + \frac{\frac{l}{2} - x_1}{\sqrt{R^2 + (\frac{l}{2} - x_1)^2}} \right)$$

$$B_2' = \frac{1}{2} B_m \left(\frac{x_2 + \frac{l}{2}}{\sqrt{R^2 + (x_2 + \frac{l}{2})^2}} - \frac{\frac{l}{2} - x_2}{\sqrt{R^2 + (\frac{l}{2} - x_2)^2}} \right)$$

$$H_1' = \frac{B_1}{\mu_0} = \frac{1}{2} n I \left(\frac{x_1 + \frac{l}{2}}{\sqrt{R^2 + (x_1 + \frac{l}{2})^2}} + \frac{\frac{l}{2} - x_1}{\sqrt{R^2 + (\frac{l}{2} - x_1)^2}} \right)$$

$$H_2' = \frac{B_2}{\mu_0} = \frac{1}{2} n I \left(\frac{x_2 + \frac{l}{2}}{\sqrt{R^2 + (x_2 + \frac{l}{2})^2}} - \frac{\frac{l}{2} - x_2}{\sqrt{R^2 + (\frac{l}{2} - x_2)^2}} \right)$$

讨论：

在外部，磁棒与载流螺线管产生 H 、 B 相同 ($B = \mu_0 H$)

在内部， $H_1' - H_1 = \frac{B_m}{\mu_0}$

载流螺线管 $m = 0$ ， $B = \mu_0 H + m = \mu_0 H$

磁棒 $J = B_m$ ， $B = \mu_0 H + J$

$\Rightarrow H_{管} \neq H_{棒}$ ， $B_{管} = B_{棒}$ 。

例 7.18.

$$H(r) = \begin{cases} \frac{I r}{2 \pi R^2} & 0 < r < R \\ \frac{I}{2 \pi r} & R < r \end{cases}$$

$$B = \begin{cases} \mu_{core} \mu_0 \frac{I r}{2 \pi R^2} & 0 < r < R \\ \mu_r \mu_0 \frac{I}{2 \pi r} & R < r < R + d \end{cases}$$

$$\mu_0 \frac{\partial \vec{H}}{\partial t} = \vec{E} + \vec{d} \times \vec{v}$$

$$(2) \mu = \frac{B}{H} = \int_0^{(M_{cur}-1)} \frac{\frac{1}{2\pi R^2}}{(M_{cur}-1)} \frac{I}{2\pi R^2} dM_{cur}$$

$$i_{in} = M_{in} \times \hat{n} = (M_{in}-1) \frac{I}{2\pi R} \quad i_{out} = M_{out} \times \hat{n} = (M_{out}-1) \frac{I}{2\pi(R+d)}$$

By 7.31.

$$\vec{i} = \vec{n} \times \hat{n} = i_0 \sin \theta, \quad i_0 = M$$

$$\begin{aligned} m &= \int_0^{\pi} i \cdot R \sin \theta \cdot 2\pi R^2 \sin^2 \theta d\theta \\ &= 2\pi M R^3 \int_0^{\pi} \sin^3 \theta d\theta \\ &= 2\pi M R^3 \int_0^{\pi} (1 - \cos^2 \theta) d(-\cos \theta) \\ &= \frac{8}{3} \pi M R^3 \end{aligned}$$

$$G_m = \vec{j} \cdot \vec{n} = G_{ms} \cos \theta \quad G_{ms} = J_0 = \mu_0 M$$

$$G_{ms} = \rho_m d \quad \leftarrow \text{假設 } \vec{E} \cdot \vec{P}_m = -\nabla \cdot \vec{P}_m = 0$$

$$\begin{aligned} P_m &= q_m d = \rho_m \cdot \frac{4}{3} \pi R^3 \cdot d \\ &= \frac{4}{3} \pi \mu_0 M R^3 \end{aligned}$$

在內部: 對稱性上, 假設 $\vec{B}$

$$\begin{aligned} B_P &= \int_0^{\pi} \frac{i_0 R \sin \theta}{4\pi \mu_0 (R^2 + x^2 - 2Rx \cos \theta)^{3/2}} \cdot R \sin \theta d\theta \\ &= \frac{M R^3}{2\mu_0} \int_0^{\pi} \frac{\sin^3 \theta d\theta}{(R^2 + x^2 - 2Rx \cos \theta)^{3/2}} \\ &= \frac{2}{3} \mu_0 M \end{aligned}$$

$$\text{在外部: } \vec{B} = \frac{1}{4\pi \mu_0 R^3} [3(\vec{p}_m \cdot \vec{r})\vec{r} - \vec{r}^2 \vec{p}_m]$$

By 7.40

$$\begin{aligned} \vec{E} &= q_m \vec{E}_+ - q_m \vec{E}_- = q_m (\vec{E}_+ - \vec{E}_-) \\ &= \rho_m \frac{\vec{H}}{\epsilon_0} \end{aligned}$$

By 7.41

$$U_m = q_m U_{m+} - q_m U_{m-}$$

$$= q_m \nabla U_m = q_m (\nabla U_m \cdot \vec{r}) = -\vec{p}_m \cdot \vec{r}$$

$$= -\frac{\vec{p}_m}{\mu_0} \cdot \vec{B} = -\vec{m} \cdot \vec{B}$$

$$m = V \cdot m = V \cdot \frac{1}{\mu_0} \vec{J} = V \cdot \frac{1}{\mu_0} \vec{P}_m = \frac{P_m}{\mu_0}$$

例 7.43

$$\vec{B} = \mu_0 \vec{H} = \frac{q_m}{4\pi} \frac{\vec{r}}{r^3}$$

$$dm = (\vec{r} \times d\vec{l}) \cdot \vec{l}_0$$

$$= \vec{r} \times (d\vec{l} \times \vec{B}) \cdot \vec{l}_0 = \vec{r} \times (d\vec{l} \times \vec{B}) \cdot \vec{l}_0$$

$$= [d\vec{l} (\vec{r} \cdot \vec{B}) - \vec{B} (\vec{r} \cdot d\vec{l})] \cdot \vec{l}_0$$

$$= \frac{q_m}{4\pi} \left(\frac{d\vec{l} \cdot \vec{r}}{r} - \frac{\vec{r} \cdot d\vec{l}}{r^3} \vec{r} \cdot \vec{l}_0 \right)$$

$$= \frac{q_m}{4\pi} \left(\frac{dr}{r} - \frac{dr}{r^3} z \right) \quad r = \frac{z}{\cos \theta}$$

$$= \frac{q_m}{4\pi} \left[d(\cos \theta) + \cos \theta \cdot \frac{dr}{r} - \frac{r dr}{r^3} \cos \theta \right]$$

$$= \frac{q_m}{4\pi} d(\cos \theta) \quad \text{对 } -q_m, q_m \rightarrow -q_m \text{ 取反}$$

$$m = \int dm_+ + dm_-$$

$$= \frac{q_m}{4\pi} (\cos \theta_{B+} - \cos \theta_{A+} - \cos \theta_{B-} + \cos \theta_{A-})$$

例 7.46

求两磁偶极子 $\vec{m}_1, \vec{m}_2$ 的相互作用力与势能

设 $\vec{m}_1$ 指向 $\vec{m}_2$ 矢径为 $\vec{r}$

① 势能法

$$U = \frac{1}{4\pi\mu_0} \left(\frac{q_{m1}}{r_{1+}} + \frac{-q_{m1}}{r_{1-}} \right) = \frac{q_{m1}}{4\pi\mu_0} \frac{r_{1-} - r_{1+}}{r_{1+} \cdot r_{1-}} \quad \vec{r}_{1+} \quad \vec{r}_{1-}$$

$$= \frac{q_{m1}}{4\pi\mu_0} \left[\left(r^2 + \frac{d^2}{4} + rd \cos \theta \right)^{-1/2} - \left(r^2 + \frac{d^2}{4} - rd \cos \theta \right)^{-1/2} \right] \quad r$$

$$= \frac{q_{m1}}{4\pi\mu_0} \left[\left(r^2 + \frac{d^2}{4} + rd \cos \theta \right)^{-1/2} - \left(r^2 + \frac{d^2}{4} - rd \cos \theta \right)^{-1/2} \right] \frac{d}{2} \cos \theta$$

$$\begin{aligned}
 & \approx \frac{q_{m1}}{4\pi\mu_0} \frac{r + rd \cos \theta}{r \cdot (r^2 - d^2 \cos^2 \theta)^{3/2}} \\
 & \approx \frac{q_{m1}}{4\pi\mu_0} \frac{r \left[1 + \frac{1}{2} \frac{d}{r} \cos \theta - \left(-\frac{d}{2r} \cos \theta \right) \right]}{r^2} \\
 & = \frac{\vec{p}_{m1} \cdot \vec{r}}{4\pi\mu_0 r^3} = \frac{\vec{m}_1 \cdot \vec{r}}{4\pi r^3}
 \end{aligned}$$

$$\begin{aligned}
 W_m &= W_{m+} + W_{m-} \\
 &= \frac{q_{m2}}{4\pi} \left[\frac{\vec{m}_1 \cdot (\vec{r} + \frac{1}{2} \vec{l}_1)}{(\vec{r} + \frac{1}{2} \vec{l}_1)^3} + \frac{\vec{m}_1 \cdot (\vec{r} - \frac{1}{2} \vec{l}_1)}{(\vec{r} - \frac{1}{2} \vec{l}_1)^3} \right] \quad \frac{d}{2} \frac{1}{r} \vec{l}_1 \\
 &\approx \frac{q_{m2}}{4\pi} \frac{\vec{m}_1 \cdot (\vec{r} + \frac{1}{2} \vec{l}_1) (r^2 - rd \cos \theta)^{3/2} - (\vec{r} - \frac{1}{2} \vec{l}_1) (r^2 + rd \cos \theta)^{3/2}}{[(r^2 + d^2)^2 - (rd \cos \theta)^2]^{3/2}} \quad \vec{r} \\
 &\approx \frac{q_{m2} \vec{m}_1}{4\pi r^6} \cdot \left[(\vec{r} + \frac{1}{2} \vec{l}_1) r^3 \left(1 - \frac{3}{2} \frac{d}{r} \cos \theta \right) - (\vec{r} - \frac{1}{2} \vec{l}_1) r^3 \left(1 + \frac{3}{2} \frac{d}{r} \cos \theta \right) \right] \\
 &= \frac{q_{m2} \vec{m}_1}{4\pi r^4} \cdot \left[\vec{r} [-3d \cos \theta] + \vec{l}_2 r \right] \\
 &= \frac{1}{4\pi r^4} (\vec{m}_1 \cdot \vec{p}_{m2} r - 3 \vec{m}_1 \cdot \vec{r} p_{m2} \cos \theta) \\
 &= \frac{\mu_0}{4\pi r^4} \left[(\vec{m}_1 \cdot \vec{m}_2) r - 3 (\vec{m}_1 \cdot \vec{r}) (\vec{m}_2 \cdot \vec{r}) \cdot \frac{1}{r} \right]
 \end{aligned}$$

$$\begin{aligned}
 \vec{J}_m &= -\nabla W_m \\
 &= -\frac{\mu_0}{4\pi} \nabla \left[(\vec{m}_1 \cdot \vec{m}_2) - 3 \frac{(\vec{m}_1 \cdot \vec{r})(\vec{m}_2 \cdot \vec{r})}{r^2} \right] \\
 &= -\frac{\mu_0}{4\pi} \left[(\vec{m}_1 \cdot \vec{m}_2) \cdot \left(-\frac{3}{r^4} \vec{r} \right) - 3 \frac{(\vec{m}_1 \cdot \vec{r})(\vec{m}_2 \cdot \vec{r})}{r^5} \left(-\frac{5}{r^3} \vec{r} \right) \right. \\
 &\quad \left. - \frac{3}{r^5} (\vec{m}_2 \cdot \vec{r}) \nabla (\vec{m}_1 \cdot \vec{r}) - \frac{3}{r^5} (\vec{m}_1 \cdot \vec{r}) \nabla (\vec{m}_2 \cdot \vec{r}) \right]
 \end{aligned}$$

$$\begin{aligned}
 \nabla (\vec{m} \cdot \vec{r}) &= \left(\frac{\partial}{\partial r} \hat{r} + \frac{1}{r} \frac{\partial}{\partial \theta} \hat{\theta} \right) (mr \cos \theta) \\
 &= m \cos \theta \hat{r} - m \sin \theta \hat{\theta} \quad \vec{r} \\
 &= \vec{m} \quad \theta' m
 \end{aligned}$$

$$\begin{aligned}
 &= \frac{3\mu_0}{4\pi} \left\{ \left[\frac{(\vec{m}_1 \cdot \vec{m}_2)}{r^4} - 3 \frac{(\vec{m}_1 \cdot \vec{r})(\vec{m}_2 \cdot \vec{r})}{r^6} \right] \vec{r} \right. \\
 &\quad \left. + \frac{1}{r^5} [\vec{m}_1 (\vec{m}_2 \cdot \vec{r}) + \vec{m}_2 (\vec{m}_1 \cdot \vec{r})] \right\} \\
 &= \frac{3\mu_0}{4\pi r^5} \left\{ [(\vec{m}_1 \cdot \vec{m}_2) r - 3 (\vec{m}_1 \cdot \vec{r})(\vec{m}_2 \cdot \vec{r})] \vec{r} + \vec{m}_1 (\vec{m}_2 \cdot \vec{r}) + \vec{m}_2 (\vec{m}_1 \cdot \vec{r}) \right\}
 \end{aligned}$$